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| **Fiscal Year:** | FY 2006 | **Task Last Updated:** | FY 11/22/2005 |
| **PI Name:** | Pitz, Robert Wendell Ph.D. | | |
| **Project Title:** | CURVED LAMINAR FLAMES IN MICROGRAVITY | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | COMBUSTION SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | COMBUSTION SCIENCE--Gaseous diffusion flames | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 37235-0001 | **Congressional District:** | 5 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2002 Physical Sciences 02-OBPR-03 |
| **Start Date:** | 10/01/2003 | **End Date:** | 09/30/2007 |
| **No. of Post Docs:** | 0 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA GRC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received NCE to 9/30/2007, per N. Hall (9/06) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNC04AA14A | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | A cylindrical burner will be studied under microgravity conditions in both the 2.2 second drop tower and the aircraft KC135. Diagnostic techniques under microgravity condition will include color video, (CH* and OH*) chemiluminescent imaging, and the Rainbow Schlieren method. Normal gravity investigations will also be performed at Vanderbilt University and will include detailed measurements of flame structure using combustion laser diagnostics. Three different kinds of fuels will be investigated in order to examine the effects of varying Lewis number. Task Objective: 1. Examine the effects of curvature on the flame structure of laminar diffusion flames. 2. Investigate the effects of curvature on the extinction limit of laminar diffusion flames. 3. Improve either reaction rate data or transport data for flames with a high degree of curvature. Practical combustion devices inevitably involve turbulence to some extent, which is characterized by the concurrence of stretch and curvature. In certain circumstances, curvature may be the | | |

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| **Task Description:** | dominant factor of some combustion phenomena, taking the extinction of the weak turbulent diffusion flame as an example. The unique geometry of the Vanderbilt cylindrical burner enables the separation of stretch effects and curvature effects, and therefore makes the investigation of these combustion phenomena possible. A comparison of the extinction stretch rate between counterflow flames and tubular flames can reveal the effects of curvature on the extinction limit. Assisted by numerical investigations, this result will reveal improved ways to prevent or even extinguish fire under microgravity. New insights to geometry-related fire safety concerns might also be introduced.<br><br>A recent numerical study showed the non-monotonous dependence on the stretch rate of the temperature of a negatively curved laminar flame with Lewis number less than one. Such phenomena can only be seen at very low stretch rate, where the gravity effects will compromise or even prevent the discovery of this phenomenon. Only under microgravity condition, such a flame can be established and thus make the study of such kind of curvature effects feasible. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Understand effect of flame curvature on flame extinction to aid in control of fires. |
| **Task Progress:** | • The PI team has finished construction of the ignitor assembly. Due to the accessibility to the ignition region, the ignitor needs to have the ability to move back and forth approximately 3 inches in a controllable manner. The desired control mechanism is the following: whenever the lever hits a limit switch, an external circuit will flip the polarity of the power supply. This results in a reversed moving. The timing to start each move is also controllable via this external circuit.<br>• The PI team has finished construction and testing of the thermocouple assembly. Thermocouple mounted on a mini-translation stage is intended to acquire a temperature profile in the radial direction during the drop. This translation stage will store a program that upon triggering can execute a sequence of actions to make the measurement (e.g. to move 100 ?m per step for 100 steps).<br><br>• NASA GRC finished the construction of the experimental rig. All the parts are mounted in position. The electrical systems are wired and past the initial testing. The rig was shipped to Vanderbilt University for testing on Sep. 14, 2005. An updated version of the electrical schematic was supplied by technician at GRC.<br><br>• The PI team finished setting up the Raman scattering experimental system for a detailed laser diagnostics on the curved laminar flames under normal gravity. The tubular burner is tested at the same time. Non-premixed tubular flames of hydrogen-air will first be studied by this non-intrusive technique (Hu, et al.).<br><br>• The PI team is conducting the necessary test on the drop rig. This includes testing (1) the flow system, (2) the electrical control system, (3) the camera system, (4) the thermocouple translation stage. Some minor upgrade is carrying out by the PI team, which includes:<br><br>1> Replace the pressure relief valves with ASME VIII certified Kunkle relief valves.<br><br>2> Install extension wire for the thermocouple probe used.<br><br>3> Install resistance wire on the ignitor assembly.<br><br>• The PI team is working closely with technical monitor at NASA GRC in preparing the pneumatic pressure test request as well as the safety review for the drop rig. The documents for the pressure test are ready. |
| **Bibliography Type:** | Description: (Last Updated: 02/18/2011) |
| **Papers from Meeting Proceedings** | Hu S, Wang P, Pitz RW. "Experimental and Numerical Investigation of Non-Premixed Tubular Flames." 44th AIAA Aerospace Sciences Meeting and Exhibit, Reno, NV, January 9-12, 2006.<br>44th AIAA Aerospace Sciences Meeting and Exhibit, Reno, NV, January 9-12, 2006. AIAA-2006-1133 paper. , Jan-2006 |